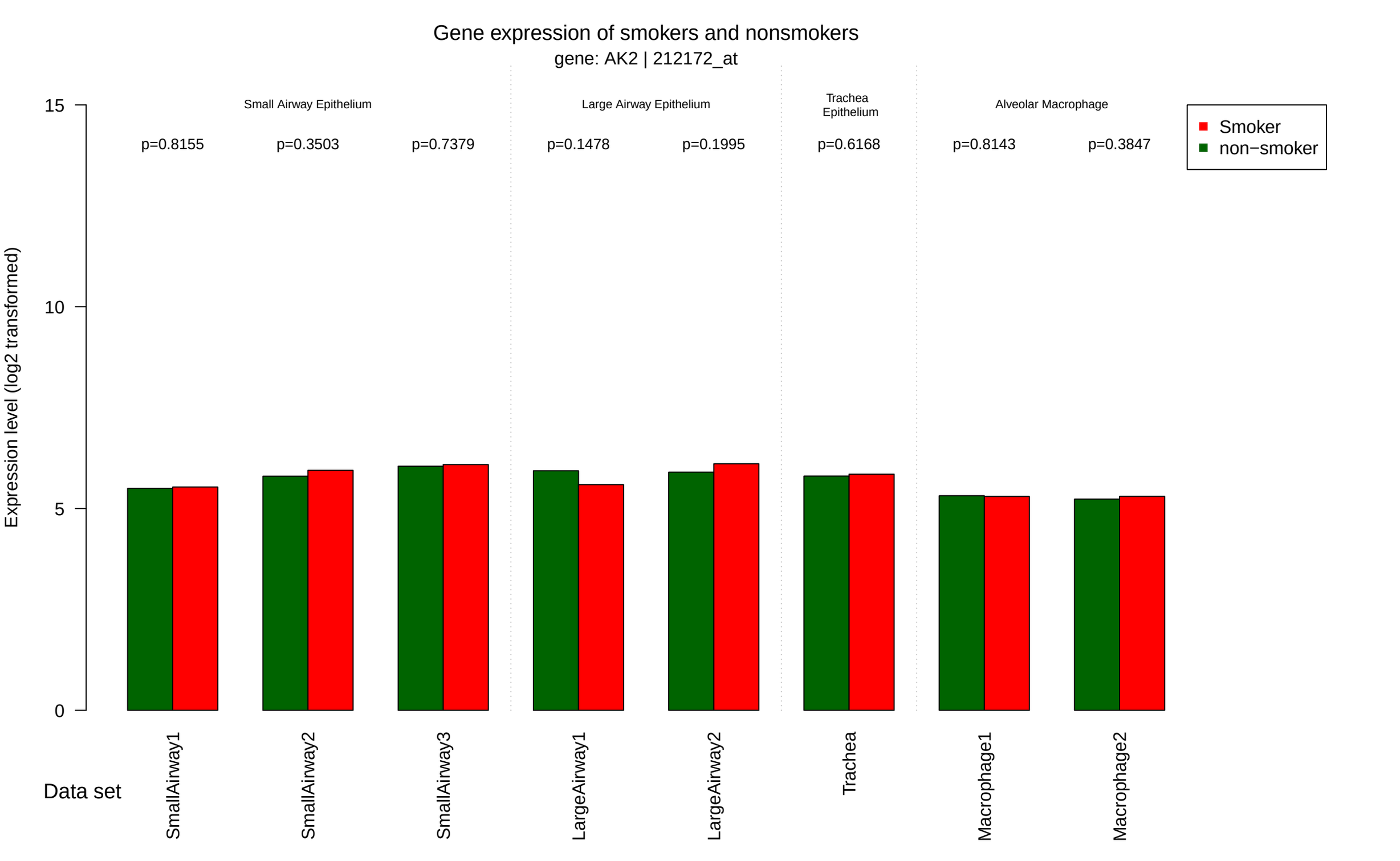
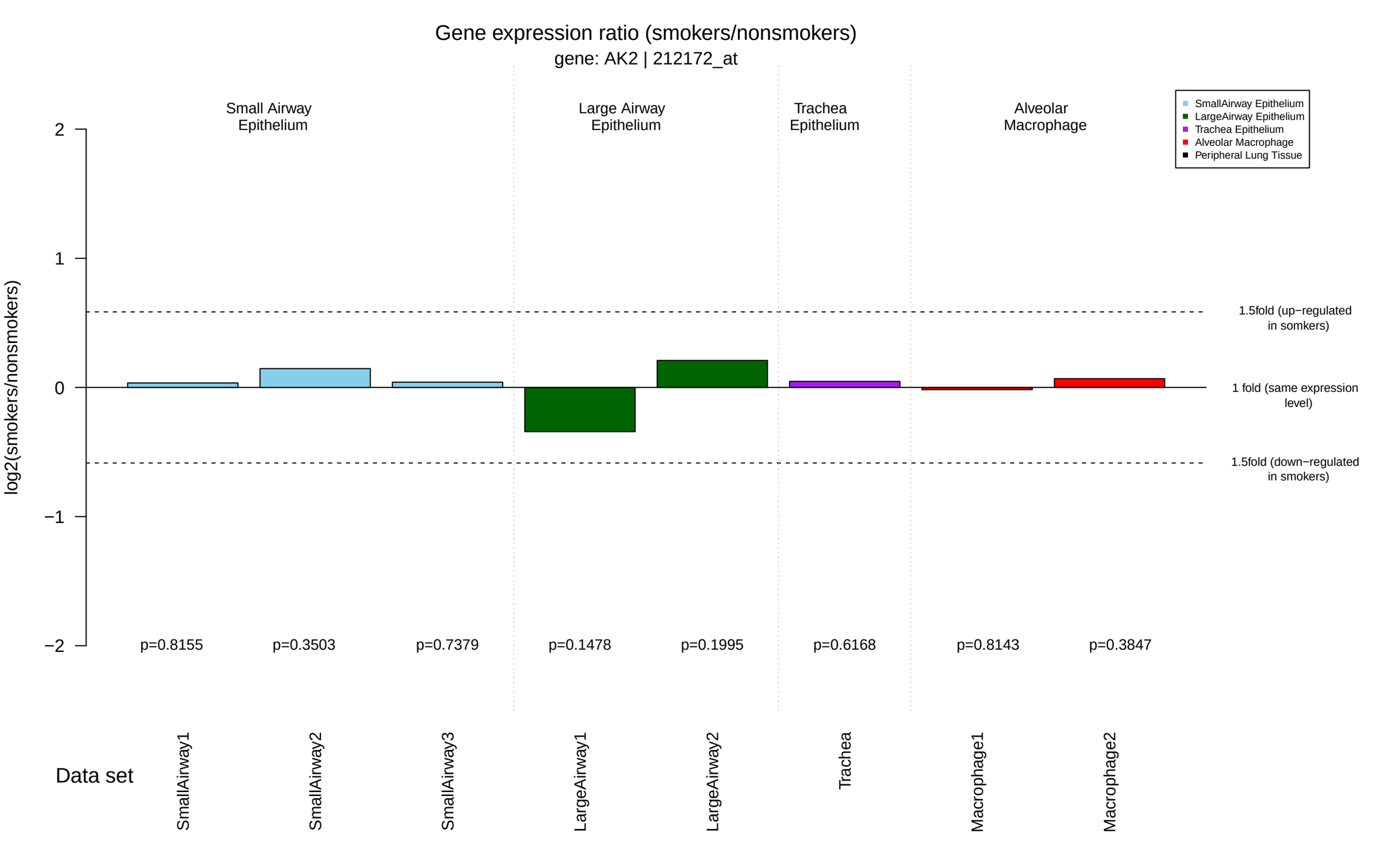
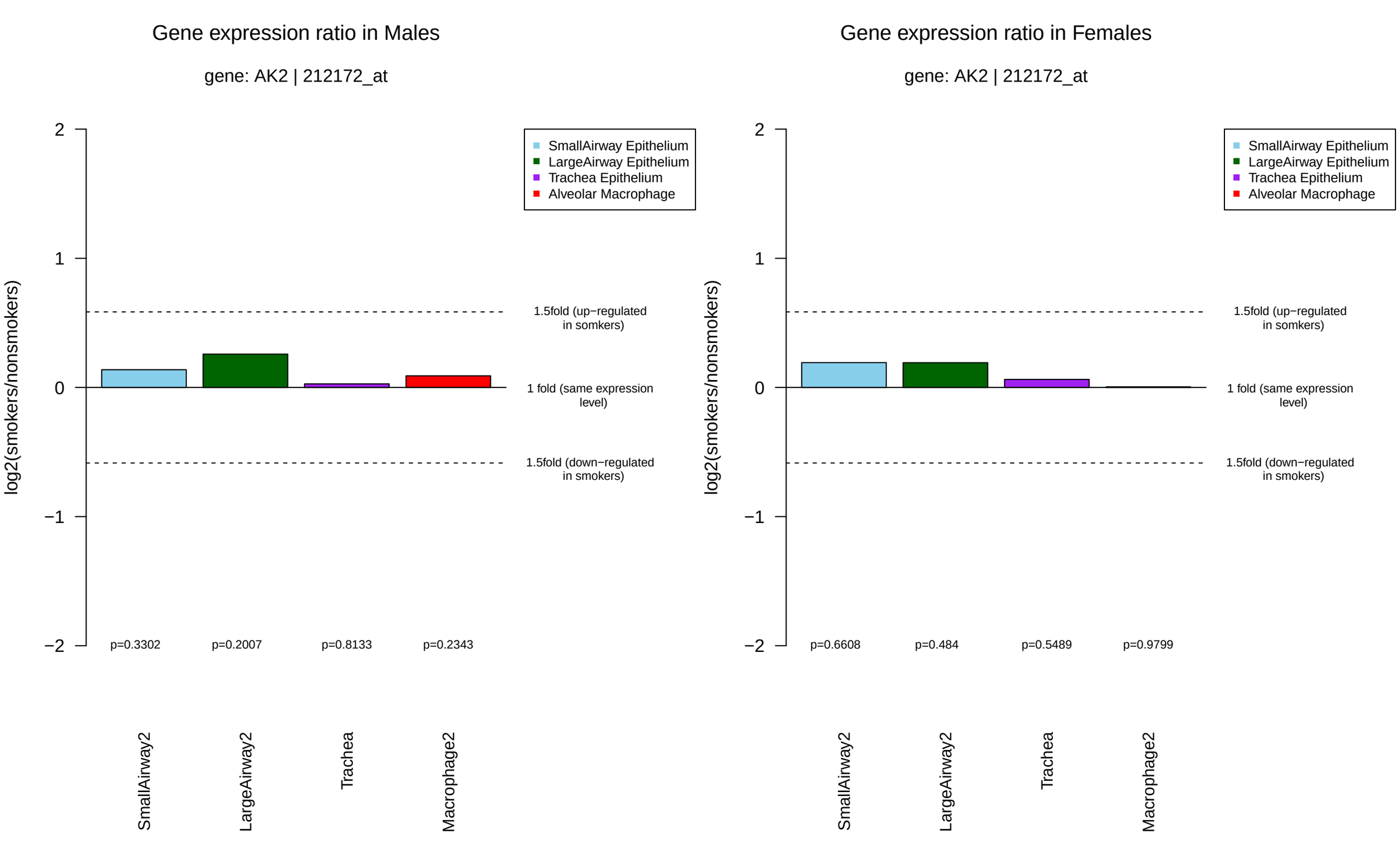
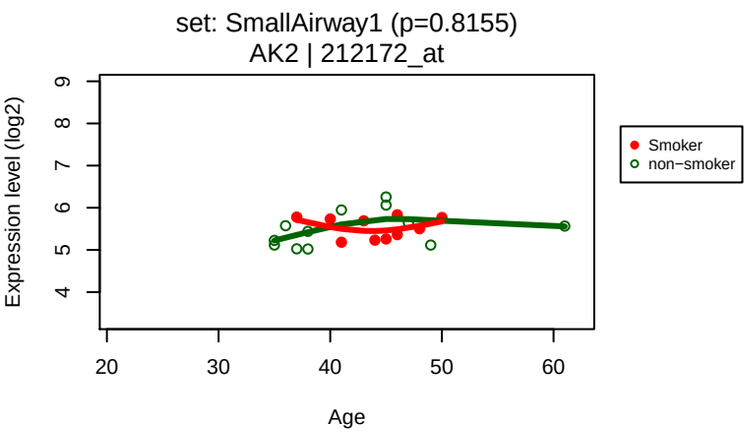
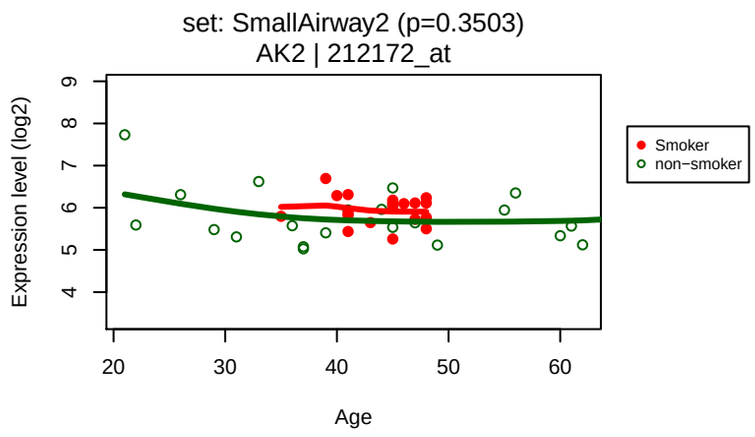
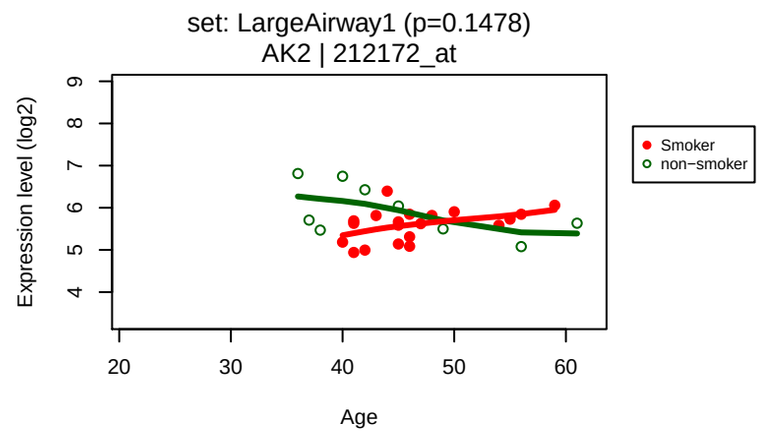

Gene expression of smokers and nonsmokers
gene: AK2 | 212172_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8155
p=0.3503
p=0.7379
p=0.1478
p=0.1995
p=0.6168
p=0.8143
p=0.3847
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

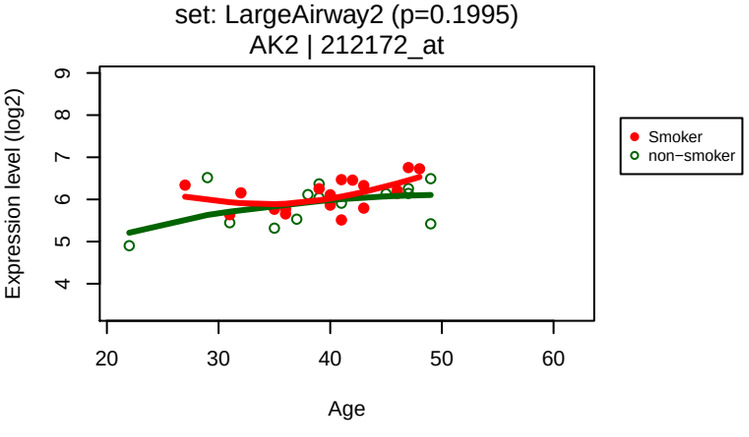

Gene expression ratio (smokers/nonsmokers)
gene: AK2 | 212172_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8155
p=0.3503
p=0.7379
p=0.1478
p=0.1995
p=0.6168
p=0.8143
p=0.3847
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

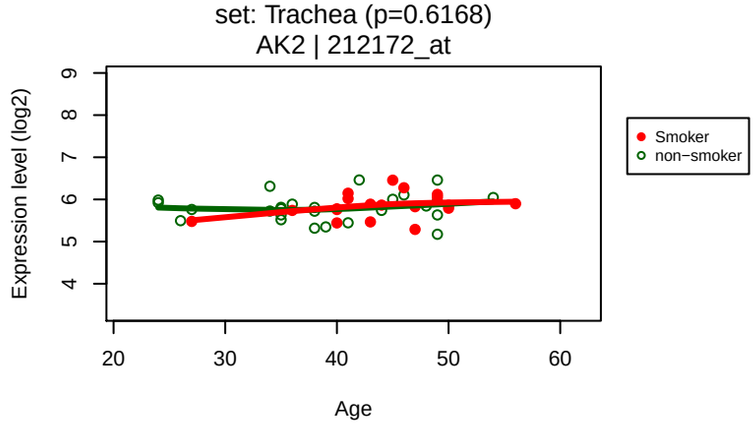

Gene expression ratio in Males
gene: AK2 | 212172_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3302
p=0.2007
p=0.8133
p=0.2343
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AK2 | 212172_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6608
p=0.484
p=0.5489
p=0.9799
SmallAirway2
LargeAirway2
Trachea
Macrophage2

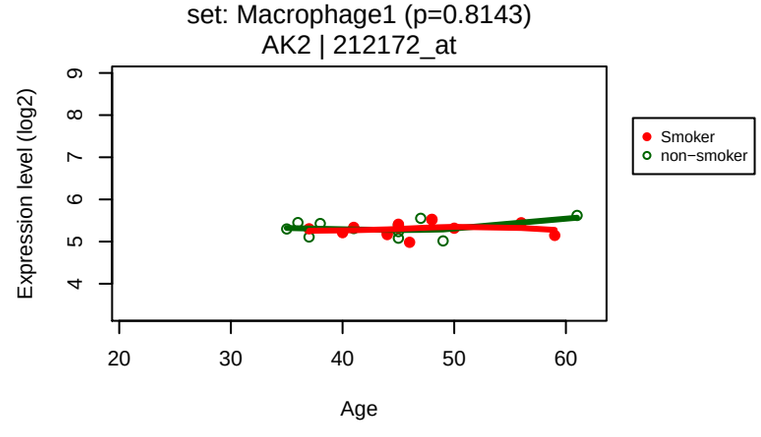

set: SmallAirway1 (p=0.8155)
AK2 | 212172_at
Expression level (log2)
Age
Smoker
non−smoker

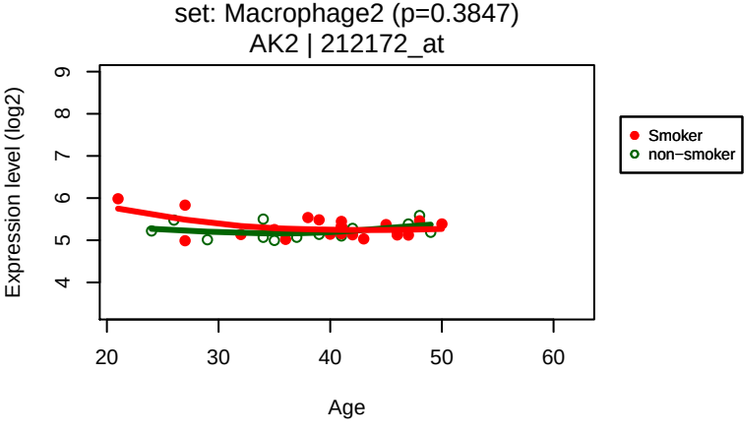

set: SmallAirway2 (p=0.3503)
AK2 | 212172_at
Expression level (log2)
Age
Smoker
non−smoker

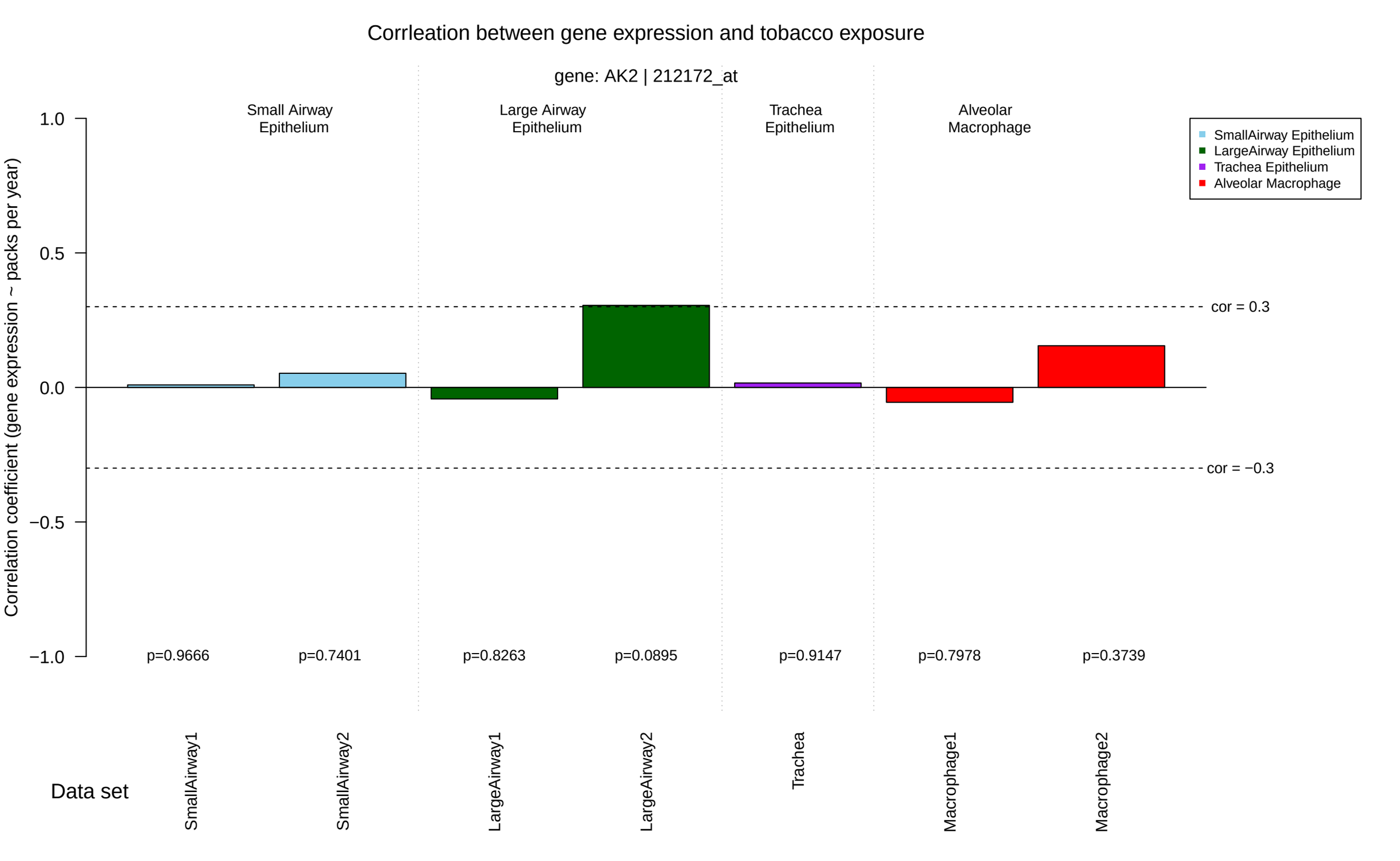

set: LargeAirway1 (p=0.1478)
AK2 | 212172_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1995)
AK2 | 212172_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6168)
AK2 | 212172_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8143)
AK2 | 212172_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3847)
AK2 | 212172_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AK2 | 212172_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9666
p=0.7401
p=0.8263
p=0.0895
p=0.9147
p=0.7978
p=0.3739
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2